

Solution to Code Wheel

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Puzzle # 1

Code: Wklv vhwqhgh lv vklwhg eb wkuhh. Fdhvdu flskhuv zhuh vxssrvhgob xvhg eb Mxolxv Fdhvdu. Wklv phwkrq lv hdvlob ghflskhuhg eb euxwh iruflqj

Answer: This sentence is shifted by three. Caesar ciphers were supposedly used by Julius Caesar. This method is easily deciphered by brute forcing.

Explanation: While you can brute force this problem by shifting each letter until you find actual words, there's an easier way to do this. Knowing that "E" is the most used letter in the alphabet, followed closely by "T," "A," "I," and "O," if you notice the most used letters in the prompt, you can figure out how much the sentence has been shifted by. There's a lot of "H"s used in the prompt, which could possibly be "E." The letters are three spaces apart, so try shifting the entire sentence back by three. If this did not work, repeat the process with a different letter.

Puzzle # 2

Code: Olyl'z huvaoly zptpsly spulhy zopma, iba dpao h zopma vm zlclu.

Answer: Here's another similar linear shift, but with a shift of seven.

Explanation: This problem uses the same steps used to solve the first puzzle. Find the most used letters in the sentence and substitute another common letter to replace them.

Puzzle # 3

Code: Skv dxplz githo atm uxjyn tcvi 13 ebwr qtn.

Hint: This sentence gets shifted by a linear function $y = mx + b$.
Undo the linear shift, and you get the original sentence.

Answer: The quick brown fox jumps over 13 lazy dogs.

Explanation: Assigning letters to 1-26 and using the equation for a line $ax+b$ will give you the shift. To undo it, use the modulo function $(ax+b) \bmod 26$. Working backwards will give you $a = 5$ and $b = 1$.

Puzzle # 4

Code: Gngjyg vmcjvp xg e Pcvqpk gvrycg.

Answer: Caesar cipher in a Caesar cipher.

Explanation: If you assembled a table of Caesar ciphers, with letters getting shifted in increments of one, you can assign letters according to a certain chosen keyword. For this puzzle, the keyword is “encrypt.” If this were a regular linear shift cipher, the repetition of characters would easily lend itself into being deciphered.

Puzzle # 5

Code: 64 25 144 144 225 400 64 25 324 25

Answer: hello there

Explanation: This one is a bit weird to catch on, but realize that these numbers are perfect squares and are associated with numbers. Take the square root of these numbers and assign them to letters until the sequence makes sense. Other variations and mathematical operations such as addition, subtraction, or division could be used to further complicate this cipher.